

1. First, we define the 3-form α as

$$\alpha := i \star [f^* dg - df^* g].$$

Next, we consider a spacetime volume $V = \Sigma \times [t_1, t_2]$. Applying Stokes' theorem and assuming that spatial boundary terms vanish, we obtain

$$\int_V d\alpha = \int_{\Sigma_2} \alpha - \int_{\Sigma_1} \alpha.$$

For $d\alpha$, we compute

$$d\alpha = id \star [f^* dg - df^* g] = i[f^* \square g - g \square f^*] d^4 V.$$

Thus, if f and g satisfy the Klein-Gordon equation, we have

$$[f^* \square g - g \square f^*] = f^* m^2 g - g m^2 f^* = 0,$$

which implies

$$\int_{\Sigma_2} \alpha - \int_{\Sigma_1} \alpha = 0.$$

Consequently, the integral

$$\int_{\Sigma} \alpha$$

is independent of the choice of Σ .

Conversely, if

$$\int_{\Sigma} \alpha$$

is independent of Σ , then

$$[f^* \square g - g \square f^*] = 0.$$

Rearranging this equation, we obtain

$$\frac{\square g}{g} = \frac{\square f^*}{f^*}.$$

Since we are considering the most general solution, and the left-hand side depends only on g , while the right-hand side depends only on f^* , the equation must be independent of both f and g . This implies that both sides must be equal to a constant, which we denote as m^2 , yielding

$$(\square - m^2)f = 0 \quad \text{and} \quad (\square - m^2)g = 0.$$

[2.] First, consider positive energy solutions $f(x) = e^{-ipx}$ and $g(x) = e^{-iqx}$. At $t = 0$:

$$\begin{aligned} f(\mathbf{x}, 0) &= e^{i\mathbf{p}\cdot\mathbf{x}}, & \partial_t f|_{t=0} &= -iE_p e^{i\mathbf{p}\cdot\mathbf{x}}, \\ g(\mathbf{x}, 0) &= e^{i\mathbf{q}\cdot\mathbf{x}}, & \partial_t g|_{t=0} &= -iE_q e^{i\mathbf{q}\cdot\mathbf{x}}. \end{aligned}$$

For the Klein-Gordon product we get:

$$\begin{aligned} (e^{-ipx}, e^{-iqx}) &= i \int [e^{i\mathbf{p}\cdot\mathbf{x}}(-iE_q e^{i\mathbf{q}\cdot\mathbf{x}}) - (iE_p e^{-i\mathbf{p}\cdot\mathbf{x}})e^{i\mathbf{q}\cdot\mathbf{x}}] d^3\mathbf{x} \\ &= (E_p + E_q) \int e^{i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}} d^3\mathbf{x} = (E_p + E_q)(2\pi)^3 \delta^{(3)}(\mathbf{q} - \mathbf{p}) \\ &= 2E_p(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}) = \tilde{\delta}(\mathbf{p} - \mathbf{q}) \end{aligned}$$

Analogously for the negative solution we obtain:

$$\begin{aligned} (e^{ipx}, e^{iqx}) &= i \int [e^{-i\mathbf{p}\cdot\mathbf{x}}(iE_q e^{-i\mathbf{q}\cdot\mathbf{x}}) - (-iE_p e^{i\mathbf{p}\cdot\mathbf{x}})e^{-i\mathbf{q}\cdot\mathbf{x}}] d^3\mathbf{x} \\ &= -(E_p + E_q) \int e^{-i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}} d^3\mathbf{x} \\ &= -2E_p(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}) = -\tilde{\delta}(\mathbf{p} - \mathbf{q}) \end{aligned}$$

And in the case of mixed solutions:

$$\begin{aligned} (e^{-ipx}, e^{iqx}) &= i \int [e^{i\mathbf{p}\cdot\mathbf{x}}(iE_q e^{-i\mathbf{q}\cdot\mathbf{x}}) - (iE_p e^{-i\mathbf{p}\cdot\mathbf{x}})e^{-i\mathbf{q}\cdot\mathbf{x}}] d^3\mathbf{x} \\ &= (E_p - E_q) \int e^{-i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}} d^3\mathbf{x} \\ &= (E_p - E_q)(2\pi)^3 \delta^{(3)}(\mathbf{p} + \mathbf{q}) \\ &= 0 \end{aligned}$$

[3.] Using the linearity of the Klein-Gordon product and the results from previous task we can write

$$\begin{aligned} (e^{ipx}, \Psi(x)) &= \sum_s \int \tilde{d}q [u(q, s)b(q, s)(e^{ipx}, e^{-iqx}) + v(q, s)d^\dagger(q, s)(e^{ipx}, e^{iqx})] \\ &= \sum_s \int \tilde{d}q [u(q, s)b(q, s)0 - v(q, s)d^\dagger(q, s)\tilde{\delta}(\mathbf{p} - \mathbf{q})] \end{aligned}$$

$$= - \sum_s v(p, s) d^\dagger(p, s)$$

and analogously for the negative solution:

$$\begin{aligned} (e^{-ipx}, \Psi(x)) &= \sum_s \int \tilde{d}q \left[u(q, s) b(q, s) (e^{-ipx}, e^{-iqx}) + v(q, s) d^\dagger(q, s) (e^{-ipx}, e^{iqx}) \right] \\ &= \sum_s \int \tilde{d}q \left[u(q, s) b(q, s) \tilde{\delta}(\mathbf{p} - \mathbf{q}) + v(q, s) d^\dagger(q, s) 0 \right] \\ &= \sum_s u(p, s) b(p, s) \end{aligned}$$

[4.] In this task, we can use the following orthogonality relations:

$$\begin{aligned} \bar{u}(p, s) u(p, s') &= 2m \delta_{ss'}, & \bar{v}(p, s) v(p, s') &= -2m \delta_{ss'}, \\ \bar{u}(p, s) v(p, s') &= 0, & \bar{v}(p, s) u(p, s') &= 0, \end{aligned}$$

which can be derived in the rest frame. For example, one obtains

$$\bar{u}(\kappa, s) u(\kappa, s') = m \begin{pmatrix} \chi_s^\dagger & \chi_s^\dagger \end{pmatrix} \gamma^0 \begin{pmatrix} \chi_{s'} \\ \chi_{s'} \end{pmatrix} = 2m \chi_s^\dagger \chi_{s'} = 2m \delta_{ss'},$$

and analogously for the other relations. Then we use the fact that these bilinears are Lorentz scalars.

Using these relations, one obtains the projection formulas

$$\begin{aligned} b(p, s) &= \frac{1}{2m} \bar{u}(p, s) (e^{-ipx}, \Psi), \\ d^\dagger(p, s) &= \frac{1}{2m} \bar{v}(p, s) (e^{ipx}, \Psi). \end{aligned}$$

In addition, we can express the creation/annihilation operators without explicitly using the Klein–Gordon product. We have

$$\begin{aligned} (e^{-ipx}, \Psi) &= i \int_{t_0} d^3\mathbf{x} \left[e^{ipx} \partial_t \Psi - (\partial_t e^{ipx}) \Psi \right] \\ &= -i \int_{t_0} d^3\mathbf{x} \left[2 (\partial_t e^{ipx}) \Psi - \partial_t (e^{ipx} \Psi) \right] \\ &= -i \int_{t_0} d^3\mathbf{x} (2iE_p e^{ipx} \Psi) = \int_{t_0} d^3\mathbf{x} (2E_p e^{ipx} \Psi), \end{aligned}$$

and similarly

$$\begin{aligned}(e^{ipx}, \Psi) &= i \int_{t_0} d^3\mathbf{x} \left[e^{-ipx} \partial_t \Psi - (\partial_t e^{-ipx}) \Psi \right] \\ &= -i \int_{t_0} d^3\mathbf{x} \left[2 (\partial_t e^{-ipx}) \Psi - \partial_t (e^{-ipx} \Psi) \right] \\ &= -i \int_{t_0} d^3\mathbf{x} (-2iE_p e^{-ipx} \Psi) = - \int_{t_0} d^3\mathbf{x} (2E_p e^{-ipx} \Psi),\end{aligned}$$

where the second term vanishes because it can be written as a total four-divergence, which gives no contribution due to boundary conditions.

Using the transformed form of the orthogonality relations,

$$\begin{aligned}\bar{u}(p, s) \gamma^0 u(p, s') &= 2E_p \delta_{ss'}, & \bar{v}(p, s) \gamma^0 v(p, s') &= 2E_p \delta_{ss'}, \\ \bar{u}(p, s) \gamma^0 v(p, s') &= 0, & \bar{v}(p, s) \gamma^0 u(p, s') &= 0,\end{aligned}$$

which follow from the Dirac equation for the wavefunctions, we can rewrite the projection formulas for the annihilation/creation operators as

$$\begin{aligned}b(p, s) &= \int_{t_0} d^3\mathbf{x} e^{ipx} \bar{u}(p, s) \gamma^0 \Psi, \\ d^\dagger(p, s) &= \int_{t_0} d^3\mathbf{x} e^{-ipx} \bar{v}(p, s) \gamma^0 \Psi.\end{aligned}$$

By complex conjugation we obtain the relations for the two complementary operators

$$\begin{aligned}b^\dagger(p, s) &= \int_{t_0} d^3\mathbf{x} e^{-ipx} \bar{\Psi} \gamma^0 u(p, s), \\ d(p, s) &= \int_{t_0} d^3\mathbf{x} e^{ipx} \bar{\Psi} \gamma^0 v(p, s).\end{aligned}$$